

Selected Exercise Solutions to Dixon's Analytic Pro-p Groups

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1 Profinite groups and pro-p groups

Exercise 1.2

Let G be a compact Hausdorff topological group. Show that G is pro-finite if and only if G is totally disconnected. ie each connected component has just one element.

Proof. Suppose G is profinite, then all open subgroups of G intersect at $\{1\}$. Therefore, G is totally disconnected. Why is the intersection of all open subgroups of G is $\{1\}$? Given a non-trivial element $g \in G$, there exists an open subset V containing 1 such that $g \notin V$ by the Hausdorff property of G . Then by the basis property, we know that there exists an open subgroup $U \subset V$. Hence there exists an open subgroup U not containing g . It follows that $\bigcap_{U \leq_o G} U = \{1\}$.

Conversely suppose G is a compact Hausdorff topological group that is totally disconnected. We prove that G is the inverse limit of finite discrete spaces. Let I be the set of equivalence relations $R \subset G \times G$ on G such that G/R is finite and discrete in the quotient topology. I is partially ordered by inclusion and is a directed set since if $R_1, R_2 \in I$ then so is $R_1 \cap R_2$ as $G/(R_1 \cap R_2)$ is also finite and discrete in the quotient topology. Then we claim that the natural projection map $\phi : G \rightarrow \varprojlim_{R \in I} (G/R)$ is a homeomorphism. Let $x = (x_R)_{R \in I} \in \varprojlim_{R \in I} (G/R)$. Then, since G is compact and each x_R is closed in G/R , $(p_R \circ \phi^{-1})(x_R)$ is compact and non-empty in G . Hence, $\bigcap_{R \in I} (p_R \circ \phi)^{-1}(x_R) = \phi^{-1}(x)$ is non-empty. So ϕ is surjective. Now, to prove injectivity, let $s, t \in G$. It suffices to find an equivalence relation $R \in I$ such that $(s, t) \notin R$. Since G is Hausdorff and totally disconnected, there exists a clopen set $U \subset G$ such that $t \in U$ and $s \notin U$. Let $R \subset G \times G$ be defined as $(x, y) \in R \iff x, y \in U$ or $x, y \notin U$. Then $G/R = \{\bar{t}, \bar{s}\}$, which is finite and is discrete. Hence $R \in I$. Hence ϕ is injective as well. ϕ is continuous, being the standard projection map. A continuous bijection is a homeomorphism hence G is a profinite group as desired. $\square$

Exercise 1.3

a) Formulate and prove an appropriate universal property for $\varprojlim G_\lambda$.

Proof. Universal Property: If R is any group and $\pi_\lambda : R \rightarrow G_\lambda$ are maps such that $\pi_\mu = \pi_{\lambda\mu} \circ \pi_\lambda$ whenever $\lambda \geq \mu$, then there exists a unique map $\varphi : R \rightarrow \varprojlim (G_\lambda)_{\lambda \in \Lambda}$ such that $\pi_\lambda = p_\lambda \circ \varphi$ for every $\lambda \in \Lambda$.

Let R be such a group. Then, set $\varphi(x) := \bigcap_{\lambda \in \Lambda} (p_\lambda^{-1} \circ \pi_\lambda)(x)$. Each subset $p_\lambda^{-1} \circ \pi_\lambda(x)$ is non-empty because if we let $g_\mu = \pi_\mu(x)$ then $(g_\mu)_{\mu \in \Lambda}$ is a vector in $\prod_\lambda G_\lambda$ that satisfies $\pi_{\lambda\mu}(g_\lambda) = g_\mu$ whenever $\lambda \geq \mu$ and by definition, $(g_\mu)_\mu \in \varprojlim G_\lambda$. The intersection contains only one vector which is exactly the vector (g_μ) constructed this way. $\square$

b) Let H be an abstract group and $\theta : H \rightarrow G$ be a homomorphism where G is a profinite group. Show that θ extends uniquely to a continuous homomorphism $\hat{\theta} : \hat{H} \rightarrow G$.

Proof. Let $\theta : H \rightarrow G$ be a homomorphism where G is a pro-finite group. Given $N \trianglelefteq G$ with finite index, we have a map $\pi_N \circ \theta : H \rightarrow G/N$. Let $N' = \ker \pi_N \circ \theta$. If $M \leq H$ is any normal subgroup of finite index, then we can find a normal subgroup $N \trianglelefteq G$ of finite index such that $N \leq \theta(M)$. Then, it follows that $N' \leq M$. Hence, $\{N'\}_{N \trianglelefteq G, [G:N] < \infty}$ forms a base for the neighborhoods of the identity in H . Therefore, we can form a profinite completion using this bases.

$$H' = \varprojlim H/N'$$

be a profinite completion. Let $\hat{H}$ denote the profinite completion. Then we automatically have a map $\pi : \hat{H} \rightarrow H'$. Now, θ induces a map from H' to G in the most natural way as follows:

$$H' \xrightarrow{\pi_{N'}} H/N' \xrightarrow{(\widehat{\pi_N \circ \theta})} G/N \xleftarrow{\pi_N} G$$

where $(\widehat{\pi_N \circ \theta})$ is induced from $\pi_N \circ \theta : H \rightarrow G/N$. Since N' is defined to be the kernel of this map, each induced map is injective. Hence given $x = (x_{N'}) \in H'$ we can define $\theta' : H' \rightarrow G$ by $\theta'(x) = ((\widehat{\pi_N \circ \theta})(x_{N'}))_N$. The maps $\pi_{N'M'} : H/N' \rightarrow H/M'$ are defined by $((\widehat{\pi_M \circ \theta}))^{-1} \circ \pi_{NM} \circ (\widehat{\pi_N \circ \theta})$. The commutativity relations are obvious and finally we set $\hat{\theta} : \hat{H} \rightarrow G$ by $\hat{\theta} = \theta' \circ \pi$. $\square$

c) Find an example of a finitely generated pro-p group G and a dense subgroup H of G , with H finitely generated as an abstract group, such that $\hat{H}_p \not\cong G$.

Proof. Take $G = \mathbb{Z}_p$ and $H = \mathbb{Z} \oplus p\mathbb{Z}$. Then, as an abstract group $H \cong \mathbb{Z}^2$ so $\hat{H}_p \cong \mathbb{Z}_p^2 \not\cong G$. $\square$

Exericse 1.5

i) Show that every closed subgroup of a procyclic group is procyclic.

Proof. Let G be procyclic and $H \leq_c G$. Then H is profinite. By proposition 1.28, $G = g^{\mathbb{Z}_p}$ for some $g \in G$. Let $f : \mathbb{Z}_p \rightarrow G$ be given by $f(\lambda) = g^\lambda$. Then, notice that if $g^\lambda \in H$, then $(g^\lambda)^{\mathbb{Z}_p} \leq H$. Now, choose i such that

$$\bigcap_{\lambda \in f^{-1}(H)} (\lambda \cdot p^i \mathbb{Z} \cap 1 \cdot p^i \mathbb{Z}) = 1 \in \mathbb{Z}/p^i \mathbb{Z}$$

$$\bigcap_{\lambda \in f^{-1}(H)} (\lambda \cdot p^i \mathbb{Z} \cap 1 \cdot p^{i+1} \mathbb{Z}) \neq 1 \in \mathbb{Z}/p^{i+1} \mathbb{Z}$$

Then $H = (g^{p^i})^{\mathbb{Z}_p}$ which proves that H is procyclic. $\square$

ii) Let G be an abelian profinite d -generated group. Prove that every closed subgroup H of G is d -generated.

Proof. We are given that

$$d = \sup_{N \trianglelefteq_o G} d(G/N)$$

Now, let $N \trianglelefteq_o G$. Then,

$$H/(H \cap N) \cong HN/N \leq G/N \implies d(H/(H \cap N)) \leq d(G/N) \leq d$$

Then, as H is a profinite group,

$$d(H) = \sup_{N \trianglelefteq_o G} d(H/(H \cap N)) \leq d$$

Since G is abelian, every subgroup of G is normal in G hence all normal subgroups of H are exactly subgroups of G contained in H , ie subgroups of H . Therefore, $H \cap N$ as N goes through all normal subgroups of G exhausts all normal subgroups of H . $\square$

iii) Let $X = \{x_1, \dots, x_m\}$ be a generating set for G , $d < m$. Then, $G/\Phi(G)$ is an elementary abelian p -group of dimension d . That is, $G/\Phi(G) \cong \mathbb{F}_p^d$ so we can consider the image of X in this quotient. Let $\overline{X}$ denote this image. By proposition 1.9, $\overline{X}$ generates $G/\Phi(G)$ so we can choose d elements in $\overline{X}$ that each fall under a different copy of $\mathbb{F}_p$ inside $G/\Phi(G)$ and hence generate $G/\Phi(G)$ as a whole. Then the preimage of these d points is a subset of X that generates G .

Exercise 1.6

i) Let $\tau : G \times \cdots \times G \rightarrow G$ be given by $\tau(x_1, \dots, x_n) = w(x_1, \dots, x_n)$. This map is continuous there its image $w(G)$ is a closed subgroup of G .

ii) Since $w(G)$ is closed, it is a pro-finite group. Let $g \in G$ be such that for every $N \trianglelefteq_o G$, there exists $w_N \in w(G)$ such that $g \equiv w_N \pmod{N}$. Then, g is the limit of the sequence $(w_N)_{N \trianglelefteq_o G}$. Since $w(G)$ is a profinite group, this proves that $g \in w(G)$. That is, there exists $x_1, \dots, x_n$ such that

$$g = w(x_1, \dots, x_n)$$

If $w(X) = X^m$ then we see that if g is an m -th power in every normal open subgroup of G , then g is an m -th power in G .

Exercise 1.7

i) Let A be a subgroup of a profinite group G . Prove that $C_G(A)$ is closed in G .

Proof. If $f, g : X \rightarrow Y$ are continuous maps and Y is Hausdorff, then the set $B = \{x \in X : f(x) = g(x)\}$ is closed in X . To see this, let $x \notin B$. Then, let U, V be disjoint open subsets of Y that contain $f(x)$ and $g(x)$ respectively. It follows that $f^{-1}(U) \cap g^{-1}(V) \not\subset B$ is an open subset that is contained in the complement of B . Hence B is closed. Now, apply this argument to maps $r_s, l_s : G \rightarrow G$ given by $r_s(g) = gs$, $l_s(g) = sg$ to conclude that for a fixed s ,

$$C_s(G) = \{g \in G : r_s(g) = l_s(g)\} = \{g \in G : gs = sg\}$$

is closed in G . Arbitrary intersections of closed sets are closed so

$$C_G(A) = \bigcap_{s \in A} C_G(s)$$

is closed in G . □

ii) If A is abelian, prove that $\overline{A}$ is also abelian.

Proof. If A is abelian then

$$A \leq C_G(\overline{A})$$

By part i, $C_G(\overline{A})$ is a closed subset of G so it must contain the closure of A . Therefore,

$$\overline{A} \leq C_G(\overline{A})$$

and so $\overline{A}$ is abelian as well. □

Exercise 1.8

Give an example of a finitely generated profinite group G whose Frattini subgroup $\Phi(G)$ is not open.

Proof. Let $G = \widehat{\mathbb{Z}} = \varprojlim (\mathbb{Z}/n\mathbb{Z})_n$. Let M be a maximal open subgroup of $\widehat{\mathbb{Z}}$. Since this profinite group is abelian, and G/M is a finite simple group, it must be isomorphic to C_p for some prime p . Then, since we have

$$\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$$

the only possible surjection $G \rightarrow C_p$ is given by mapping $\mathbb{Z}_q$ to 1 for every $q \neq p$ and sending $(x_n)_n \in \mathbb{Z}_p$ to $x_1 \in \mathbb{Z}/p\mathbb{Z} \cong C_p$. Then the kernel of this map is a maximal open subgroup given by

$$M = p\mathbb{Z}_p \times \prod_{q \neq p} \mathbb{Z}_q$$

Therefore,

$$\Phi(\widehat{\mathbb{Z}}) = \prod_p p\mathbb{Z}_p$$

and it is not open in $\widehat{\mathbb{Z}}$ since

$$\widehat{\mathbb{Z}}/\Phi(\widehat{\mathbb{Z}}) \cong \prod_p \mathbb{Z}_p / \prod_p p\mathbb{Z}_p \cong \prod_p \mathbb{F}_p$$

is an infinite group, which proves $\Phi(\widehat{\mathbb{Z}})$ does not have finite index, hence cannot be an open subgroup. Finally, the isomorphism between $\varprojlim \mathbb{Z}/n\mathbb{Z}$ and $\prod_p \mathbb{Z}_p$ is given by sending an element $(a_p)_p$ in the latter group to the element $(x_n)_n$ of the former group where

$$x_n = \prod_{p|n} (a_p \mod p^{j_p(n)})$$

where $j_p(n) \in \mathbb{N}$ is the greatest power of p in the prime decomposition of n . $\square$

Exercise 1.9

$SL_n(\mathbb{Z}/m\mathbb{Z})$ is generated by elements of the form $I + e_{ij}$ where e_{ij} is the matrix with 1 on ij -th entry and 0 otherwise and a diagonal matrix of the form $(1, -1, 1, 1, \dots, 1)$. All of these have determinant 1 both in $\mathbb{Z}$ and in $\mathbb{Z}/m\mathbb{Z}$ hence the map $SL_n(\mathbb{Z}) \rightarrow SL_n(\mathbb{Z}/m\mathbb{Z})$ is surjective. Let $(A_i)_i \in \varprojlim SL_n(\mathbb{Z}/p^i\mathbb{Z}) \cong \varprojlim SL_n(\mathbb{Z})/K(p^i)$. Then, map A_i to the element $B \in SL_n(\mathbb{Z}_p)$ where

$$B_{kl} = \sum_{i=0}^{\infty} (A_i)_{kl} \cdot p^i$$

This mapping is obviously injective. Moreover, given $B \in SL_n(\mathbb{Z}_p)$, each $B_{kl} = \sum_{i=1}^{\infty} a_i^{(kl)} \cdot p^i$ for some $a_i \in \mathbb{Z}/p^i\mathbb{Z}$. Then, the matrix $B_i := (a_i^{(kl)})_{1 \leq k, l \leq n}$ is in $SL_n(\mathbb{Z}/p^i\mathbb{Z})$ so the element $(B_i)_i$ is mapped to B under this map. Hence, the given map is also surjective. A similar construction also proves that

$$\varprojlim SL_n(\mathbb{Z})/K(m) \cong \varprojlim SL_n(\widehat{\mathbb{Z}})$$

Exercise 1.10

Fix a prime p and a positive integer n . Put

$$\Gamma_j = \{g \in SL_n(\mathbb{Z}_p) : g \equiv 1_n \pmod{p^j}\}$$

1. Show that Γ_1 is pro- p .
2. Show that Γ_1 is finitely generated.

Proof. Define

$$B_m = \{(a_{ij}) \in M_n(\mathbb{Z}_p) : a_{ij} \in p^m\mathbb{Z}, a_{ii} \in 1 + p^m\mathbb{Z}\}$$

Notice that B_m is open and also closed in $M_n(\mathbb{Z}_p)$, since each $p^m\mathbb{Z}$ is both open and closed in $\mathbb{Z}_p$. Moreover,

$$\Gamma_j = SL_n(\mathbb{Z}_p) \cap B_j$$

is an open and closed subset of $SL_n(\mathbb{Z}_p)$, with the subspace topology induced from $M_n(\mathbb{Z}_p)$. Given $A \leq_o \Gamma_1$, there exists an element $A' \in A$ with the smallest $m \in \mathbb{N} - 0$ such that $A_{ij} = p^m a_{ij}$ for some $a_{ij} \in \mathbb{Z}_p$ and $1 \leq i, j \leq n$. Then $\Gamma_j \subset A \subset \Gamma_1$. So $\{\Gamma_j\}_{j=1}^{\infty}$ form a basis for the neighbourhoods of $1 \in \Gamma_1$. Direct $\{\Gamma_j\}$ by reverse inclusion and let $\pi_{mn} : \Gamma_1/\Gamma_m \rightarrow \Gamma_1/\Gamma_n$ be the projection map. If we can show that $\Gamma_n \leq \Gamma_1$ and that Γ_1/Γ_n is a finite p -group then Γ_1 is a pro- p group. To that end, consider the map

$$\Phi : \Gamma_i \rightarrow M_n(\mathbb{Z}/p\mathbb{Z}) \tag{1}$$

$$I + p^i A \mapsto A \pmod{p} \tag{2}$$

Notice that $\ker \Phi = \Gamma_{i+1}$ and therefore $\Gamma_{i+1} \leq \Gamma_i$ and Φ is a homomorphism since

$$(I + p^i A)(I + p^i B) = I + p^i(A + B + p^i AB) \mapsto A + B$$

hence Γ_i/Γ_{i+1} is isomorphic to an additive subgroup of $M_n(\mathbb{Z}/p\mathbb{Z})$, which has power p^{n^2} . So definitely Γ_i/Γ_{i+1} is a finite p -group. Finally,

$$\Gamma_1/\Gamma_j \cong \prod_{i=1}^{j-1} \Gamma_i/\Gamma_{i+1}$$

hence it is a p -group as well. This proves that

$$\Gamma_1 = \varprojlim (\Gamma_1/\Gamma_j)_{j \geq 1}$$

is a pro-p group.

As for the other part, Γ_1 is finitely generated iff $\Phi(\Gamma_1)$ is open in Γ_1 . Now, $\Phi(\Gamma_1) = \Gamma_1^p[\Gamma_1, \Gamma_1]$ and we have

$$\Gamma_1^p \leq \Gamma_2, \quad [\Gamma_1, \Gamma_1] \leq \Gamma_2$$

Hence $\Phi(\Gamma_1) \leq \Gamma_2$ and since Γ_1/Γ_2 is elementary abelian p-group every maximal subgroup of Γ_1 contains Γ_2 so

$$\Phi(\Gamma_1) = \Gamma_2 \trianglelefteq_o \Gamma_1$$

□

Exercise 1.11

Let G be a profinite group and p be a fixed prime. For $N \trianglelefteq_o G$, denote by $P(N)$ the subset of Sylow p-subgroups of G/N . Show that G has a closed subgroup P such that $PN/N \in P(N)$ for every $N \trianglelefteq_o G$. Since P is closed, it is profinite. Every open normal subgroup of P is of the form $P \cap N$ for $N \trianglelefteq_o G$ (proposition 1.2). Then, by the isomorphism theorems,

$$P/(P \cap N) \cong PN/N \in P(N)$$

shows that every quotient by open normal subgroups is a p-group. Hence P is a pro-p group. P is a maximal pro-p subgroup because if M is a pro-p subgroup of G . Then, for a given normal subgroup $N \trianglelefteq_o G$, $|MN/N| = p^\beta$ and $|PN/N| = |P_N/N| = p^\alpha$ where $\alpha \geq \beta$. Hence, a conjugate of MN/N is contained inside P_N/N . Therefore, P is maximal.

Proof. Since $N \trianglelefteq_o G$, it has finite index so $P(N)$ is a finite set and is compact with the discrete topology. Now, if $N \leq M$, we can define

$$\begin{aligned} \pi_{NM} : P(N) &\rightarrow P(M) \\ H/N &\mapsto HM/M \end{aligned}$$

We have to show HM/M is a Sylow-p subgroup of G/M . Suppose $|N| = a$, $|M| = ab$, $|G/N| = p^\alpha \ell$, and $|G/M| = p^\beta k$. Then,

$$|HM/M| = |HM|/|M| = \frac{|H| \cdot |M|}{|H \cap M| \cdot |M|} = \frac{|H|}{|H \cap M|} = \frac{p^\alpha \cdot |N|}{|H \cap M|} = p^\gamma$$

The last equality holds because $N \leq H \cap M$ so the integer must be a power of p . Now, by proposition 1.4, the inverse limit is non-empty so let $(P_N/N) \in \varprojlim P(N)$. Then

$$P_N M/M = P_M/M \quad \text{whenever } N \leq M$$

and since $M \leq P_M$, it follows that

$$P_N \leq P_M \quad \text{whenever } N \leq M$$

Then, consider the subgroup

$$P = \bigcap_{N \trianglelefteq_o G} P_N$$

Firstly, notice that

$$PN/N = \bigcap_{M \trianglelefteq_o G} P_M N/N$$

and if $M \leq N$, then $P_M N/N = P_N/N$. If $M \geq N$ then $P_M N/N \geq P_N/N$ and if M, N are not comparable, then

$$P_M N/N \geq P_{M \cap N} N/N = P_N/N$$

Hence,

$$PN/N = P_N/N \in P(N)$$

for every $N \trianglelefteq_o G$. Moreover, each P_N is open because $|G/P_N| \leq |G/N| < \infty$ and therefore each P_N is closed as well. Hence, P is closed. $\square$

Exercise 1.15

Assuming Kostrikin's theorem, let G be a finitely-generated pro- p group. Then $\pi_k(G) = \overline{G^{p^k}}$ and if $x \in G/\pi_1(G)$ then $x^p = 1$ so $G/\pi_1(G)$ is a finitely-generated group of exponent p . By Krasinski's theorem, the order of $G/\pi_1(G)$ is finite. Hence, by theorem 1.17, $\pi_1(G)$ is open in G , being a subgroup of finite index. Similarly, $\pi_k(G)$ is open in $\pi_{k-1}(G)$. Here, $\pi_{k-1}(G)$ is given the subspace topology therefore $\pi_k(G) = A \cap \pi_{k-1}(G)$ where A is an open subset of G . Then it follows that $\pi_k(G)$ is open in G as well. This proves the first direction.

For the other direction, take $G = F_d^p$ to be the free pro- p group on d generators. Then G/G^p has finite index since G^p is open by assumption. Now let H be any d -generated, exponent- p group and let $\hat{H}$ be its pro- p completion. Then let $\{s_1, \dots, s_d\}$ be the generators of H and let $\{h_1, \dots, h_d\}$ denote the generators of G . Let $f : \{h_1, \dots, h_d\} \rightarrow H$ be given by $f(h_i) = s_i$. Then by the universal property of the free pro- p group, there exists a unique continuous homomorphism $\hat{f} : G \rightarrow \hat{H}$ that sends $h_i \mapsto s_i$. This map induces a continuous homomorphism $\hat{f} : G/G^p \rightarrow \hat{H}$ since $\hat{H}$ is an exponent- p group, G^p lies in the kernel of $\hat{f}$. Additionally, we also have the inclusion map $H \hookrightarrow \hat{H}$. Hence, H can be associated with a subgroup of G/G^p . By assumption, this shows that the order of H is bounded by the order of G/G^p .

Exercise 1.16

Proof. $\Rightarrow$ direction. Let G be a pro- p group generated by d elements. Then

$$\overline{G^{p^k}} = \bigcap_{N \trianglelefteq_o G} G^{p^k} N$$

is a closed subgroup, since it is the intersection of closed sets. A closed subset of G is open if and only if it has finite index. By Zelmanov's theorem, since G/G^{p^k} is a d -generated, exponent- p^k group, its order is bounded by $f(d, p^k)$. Hence, $\overline{G^{p^k}}$ is open in G .

$\Leftarrow$ direction: Take G to be the free group on d generators. Then, by assumption $\overline{G^{p^k}}$ is open in G so $G/\overline{G^{p^k}}$ has finite order. Notice that since $\overline{G^{p^k}}$ is open, we have $G^{p^k} = \overline{G^{p^k}}$. Moreover, $G/\overline{G^{p^k}}$ is d -generated and is exponent p^k so given any other group H that is d -generated and exponent p^k , the map that sends the generators of G to the generators of H extends uniquely to a map from $G \rightarrow \hat{H}$ and it induces a map $G/\overline{G^{p^k}} \rightarrow \hat{H}$ since every element of $\hat{H}$ has order p^k , which means G^{p^k} is in the kernel. Then, it follows that the degree of H is bounded by that of $G/\overline{G^{p^k}}$. $\square$

Exercise 1.19 (Martinez 1994)

Step 1: We can write $G = \varprojlim G_\lambda$ where G_λ form an inverse system of finite p -groups. If $x = (x_\lambda)_\lambda \in G$ then $x^{p^k} = (x_\lambda^{p^k})_\lambda$. Suppose for each G_λ , $G_\lambda^{p^k} = S_\lambda^{(n)}$ where n depends on G_λ and $S_\lambda = \{x^{p^k} : x \in G_\lambda\}$. In this case, since each G_λ has the discrete topology, $S_\lambda^{(n)}$ is closed, hence so is $G_\lambda^{p^k}$. Since G^{p^k} is a subgroup of the direct product $\prod_{\lambda \in \Lambda} G_\lambda^{p^k}$, it is also closed.

Step 2: Notice that $G^{p^k} \trianglelefteq_o G$ so G^{p^k} is finitely generated by Proposition 1.7. In fact, let T be transversal to $H = G^{p^k}$ then for each $t \in T, x \in X$ there exists $s = s(t, x) \in T$ such that $Htx = Hs$. Then, the proof of proposition 1.7 shows that

$$Y = \{tx \cdot s(t, x)^{-1} : t \in T, x \in X\}$$

generates H topologically. Since G/G^{p^k} is at most d -generated, by Exercise 1.16,

$$|T| = |G/G^{p^k}| \leq f(d, p^k)$$

so

$$|H| \leq df(d, p^k)$$

Since G^{p^k} is generated by $\{g^{p^k} : g \in G\}$, by exercise 5 part iii, and is finitely generated, say r -generated, there exists an r element subset $\{g_1^{p^k}, \dots, g_r^{p^k}\}$ that generates H . Now the above part shows that $r \leq df(d, p^k)$.

Step 3: Let X denote the set of all conjugates of the elements $g_1^{p^k}, \dots, g_r^{p^k}$ in H . Since G is a finite p -group it is nilpotent, hence so is the subgroup H . Then by lemma 1.23, we can write an element of $[H, H]$ in the form

$$[x_1, g_1^{p^k}] \dots [x_r, g_r^{p^k}] = x_1 g_1^{p^k} x_1^{-1} g_1^{-p^k} \dots x_r g_r^{p^k} x_r^{-1} g_r^{-p^k} = a_1 g_1^{-p^k} \dots a_r g_r^{-p^k} \in X^{(2r)}$$

where $a_i = x_i g_i^{p^k} x_i^{-1} \in X$. Similarly, any element of X can be written as an element of $[H, H]$. Therefore, $[H, H] = X^{(2r)}$. Next, notice that $[H, H] = X^{(2r)} \subset S^{(2r)}$. Moreover, since $H/[H, H]$ is abelian, any element of $H^{p^k}/[H, H]$ is a p^k -th power. So any element of $[H, H]H^{p^k}$ can be written as $y x^{p^k}$ where $y \in [H, H]$ and $x^{p^k} \in H^{p^k}$. Thus,

$$[H, H]H^{p^k} \subset S^{(2r+1)}$$

as desired.

Step 4: Any element mod $[H, H]H^{p^k}$ is congruent to a product of $g_1^{p^k}, \dots, g_r^{p^k}$ up to maximum $(p^k - 1)$ -th power, in this order because $H/[H, H]H^{p^k}$ is abelian. Therefore,

$$H = S^{(r \cdot (p^k - 1) + 2r + 1)} = S^{(r(p^k + 1) + 1)}$$

as desired. This proves that if G is a finitely generated pro- p group then for every k , the subgroup G^{p^k} is closed in G .

Exercise 1.20

i) Let X be a finite set and $F(X)$ be the free group on X . Show that every mapping $\phi : X \rightarrow G$ where G is a pro- p group, extends uniquely to a continuous homomorphism $\phi' : \widehat{F_p} \rightarrow G$.

Proof. Given such $\phi : X \rightarrow G$, there exists a unique $\varphi : F(X) \rightarrow G$ by the universal property of the free group. Now, by exercise 1.3 part b, φ extends uniquely to a continuous homomorphism $\widehat{F(X)}_p \rightarrow G$ and this proves the statement. $\square$

ii) Suppose $Y \subseteq X$. Show that the inclusion mapping $F(Y) \rightarrow F(X)$ induces an isomorphism of $\widehat{F(Y)}_p$ onto the closed subgroup of $\widehat{F(X)}_p$ generated by the image of Y .

Proof. Firstly, note that by part i), given $Y \rightarrow \widehat{F(X)}_p$, there exists a unique continuous homomorphism $\phi : \widehat{F(Y)}_p \rightarrow \widehat{F(X)}_p$. Now let $x = (x_N) \in \widehat{F(Y)}_p$ such that $\phi(x) = 1 \in \widehat{F(X)}_p$. If $x \neq 1 \in \widehat{F(Y)}_p$ then there exists a normal subgroup N of $F(Y)$ of finite index such that $x_N \notin N$. Suppose that $N = F(Y) \cap M$ for some $M \trianglelefteq F(X)$. Then, $x_N \in M, x_N \in F(Y) \implies x_N \in N$. So it suffices to prove that for every normal subgroup N of $F(Y)$, $N = F(Y) \cap M$ for some normal subgroup M of $F(X)$. Given N , set

$$M := \{v \cdot w \cdot v^{-1} : w \in N, v \in X\}$$

Then $M \trianglelefteq F(X)$ and obviously $N = F(Y) \cap M$. Hence, the induced map ϕ is injective and the result follows. $\square$

2 Powerful p-groups

Exercise 2.2

Let $N, M \trianglelefteq G$.

- i) If N p.e. in G , then N^{p^i} p.e. in G for all i .
- ii) If $[N, M]$ p.e. N then $[N^p, M] = [N, M]^p$.
- iii) If N p.e. G and M p.e. G then $[N^{p^i}, M^{p^j}] = [N, M]^{p^{i+j}}$.

Proof. i) By proposition 2.3, N^p p.e. G . Then by induction, the result follows. What is the point of theorem 2.7?

- ii) To prove $[N^p, M] \leq [N, M]^p$, mod out by $[M, N]^p$ and so we can assume $[N, M]^p = 1$. Then, since $[N, M]$ p.e. in N ,

$$[N, M, N] \leq [N, M]^p = 1$$

so we see that for $x \in N, g \in M$, by carrying out the same calculation as in proposition 2.3, now $[N, M, N] = 1$ so it is definitely central in N , we see that

$$[x^p, g] = [x, g]^p [x, g, x]^{p(p-1)/2} = 1$$

since $[N, M]^p = 1 = [N, M, N]$. Thus, $[N^p, M] = 1$ as well so $[N^p, M] \leq [N, M]^p$.

For the other direction,

- iii) If N, M are p.e. G , then by exercise 1 part d, $[N, M]$ is powerfully embedded in N . Then by part 2,

$$\begin{aligned} [N^{p^i}, M^{p^j}] &= [N^{p^{i-1}}, M^{p^j}]^p \quad \text{IH on } i+j \\ &= ([N, M]^{p^{i+j-1}})^p \\ &= [N, M]^{p^{i+j}} \end{aligned}$$

$\square$

Exercise 2.5

Proof. i) Since G is a finitely generated powerful p-group, theorem 2.7 applies so we get a homomorphism

$$G/G^p \rightarrow G^{p^{e-1}}/G^{p^e} \cong G^{p^{e-1}} \quad x \mapsto x^{p^{e-1}}$$

This homomorphism extends to G since for $x \in G^p$, $x = g^p$ for some $g \in G$ and so $x^{p^{e-1}} = g^{p^e} = 1$ so G^p is in the kernel of this map. Hence we see that

$x \mapsto x^{p^{e-1}}$ is an endomorphism of G .

ii) Notice that

$$G/\ker \phi \cong G_e \quad \ker \phi = E_{e-1}$$

so E_{e-1} is a subgroup of G of order $[G : G_e]$

iii)

iv) Notice that each E_k is characteristic and

$$E_1 \subset E_2 \subset \cdots \subset E_{e-1} \subset E_e = G$$

If every characteristic subgroup of G is powerful, then by part i), E_{e-1} is powerful so replacing G by E_{e-1} and E_{e-1} by E_{e-2} and the map by $x \mapsto x^{p^{e-2}}$, we see that E_{e-2} is a subgroup of E_{e-1} hence also a subgroup of G . Now, E_{e-2} is also powerful by assumption. Then we can apply this argument again and again until we prove that E_k is a subgroup of G for every k . $\square$

Exercise 2.8

a) Suppose p is odd. Prove that G is powerful if and only if G is the product of $d(G)$ cyclic subgroups.

Proof. If G is powerful, then corollary 2.8 solves the $\Rightarrow$ direction. For the converse, let's write

$$G = \langle a_1 \rangle \cdots \langle a_d \rangle$$

where $d = d(G)$ and let $p^{\alpha_i} := |\langle a_i \rangle|$. Then we know that $G/\Phi(G) \cong (C_p)^d$ and that $\langle a_1 \Phi(G) \rangle \cdots \langle a_d \Phi(G) \rangle$ generates $G/\Phi(G)$ by proposition 1.9, since a finite p -group is profinite. Now notice that

$$G/G^p \cong \langle a_i^{p^{\alpha_i-1}} \rangle \cdots \langle a_d^{p^{\alpha_d-1}} \rangle$$

and since $G^p \leq \Phi(G)$, $G/G^p \geq G/\Phi(G)$. Notice that $|G/G^p| = p^d = |G/\Phi(G)|$. Therefore,

$$G/\Phi(G) = G/G^p$$

hence G/G^p is abelian. So G is powerful. $\square$

b) Show by example that this fails if $p = 2$.

Proof. Consider

$$G = D_8 = \{r, s | r^4 = s^2 = 1, srs = r^{-1}\}$$

Then we see that $G = \langle r \rangle \langle s \rangle$ and

$$G^4 = \langle r^4 \rangle \langle s^4 \rangle = 1$$

and $G/G^4 = G$ is not abelian so G is not powerful. $\square$

3 Pro-p Groups of Finite Rank

4 Uniformly Powerful Groups

Exercise 4.1

Proof.

$$a^h b^h = (ab)c(ab)c^2(ab)c^3 \dots (ab)c^{h-1}(ab) = (ab)^h c^{h(h-1)/2}$$

For the latter part, since $[y, x^{-1}] \in [G_k, G_k] \leq G_{2k}$ and $[x, [y, x^{-1}]] \in [G_k, G_{2k}] \leq G_{3k} \leq G_{k+m}$, the map $x \mapsto x^{p^n}$ is a homomorphism by the previous part.

$$(xy)^{p^n} = x^{p^n} y^{p^n} [y, x^{-1}]^{p^n(p^n-1)/2} \equiv x^{p^n} y^{p^n} \pmod{G_{k+m}}$$

□

Exercise 4.2

Proof.

$$\begin{aligned} -(y, x)_n^{p^{2n}} &= -[y^{p^n}, x^{p^n}] \\ &= [y^{p^n}, x^{p^n}]^{-1} \\ &= [x^{p^n}, y^{p^n}] \\ &= (x, y)_n^{p^{2n}} \end{aligned}$$

Hence $(x, y)_n = -(y, x)_n \implies (x, y) = -(y, x)$. Moreover, since $(x, y)_n \in G_{2n+2-2n} = G_2$ for odd p , and since G_2 is a closed subset, $(x, y) = \lim_n (x, y)_n \in G_2$ as well. □

Exercise 4.3

Proof. Write $\lambda = a + bp^n$. Then, let $c = ax$ and $d = p^n b \cdot x$ we have $d = (x^b)^{p^n} \in G_{n+1} \subset G_n$ so by Lemma 4.28 we can write

$$\begin{aligned} (\lambda x, y) &\equiv (\lambda x, y)_{n-1} \\ &\equiv (\lambda x, y)_n \\ &\equiv (cd, y)_n \\ &\equiv (c, y)_n \\ &\equiv (c, y)_{n-1} \\ &\equiv (c, y) \pmod{G_{n+1}} \end{aligned}$$

as desired. □

6 Normed Algebras

Exercise 6.1

Let $(R, \|\cdot\|)$ be a normed ring. Show that the 'closed ball' $B_\delta(a) = \{x \in R : \|x - a\| \leq \delta\}$ is open and closed.

Proof. Let $x \in B_\delta(a)$. We claim that $B_{\delta/2}(x) \subset B_\delta(a)$, which proves that $B_\delta(a)$ is open. Notice that

$$\|y - a\| = \|y - x + x - a\| \leq \max(\|y - x\|, \|x - a\|) \leq \delta$$

The set is closed since if $y \notin B_\delta(a)$, let $\gamma = \|y - a\| > \delta$, then it is easy to see that $B_{\gamma/2}(y) \cap B_\delta(a) = \emptyset$. $\square$

Exercise 6.2

Proof. i) By symmetry, it suffices to show that the concepts of a Cauchy sequence arising from the metric defined using (R'_i) is the same as that arising from the metric defined using (R_i) . So let (a_n) be a Cauchy sequence in the metric space constructed using (R_i) . Let $k \in \mathbb{N}$ be given. Notice that

$$\|a_n - a_m\| \leq c^{-k} \iff a_n - a_m \in R'_k$$

So (a_n) forms a Cauchy sequence in (R'_i) -metric if the sequence of cosets $(a_n + R'_k)_{n=1}^\infty$ stabilizes after some $n = N_k$, for every k . Now, since (R'_i) and (R_i) are cofinal, given $k \in \mathbb{N}$, there exists $k(j) \in \mathbb{N}$ such that $R'_k \supset R_{k(j)}$. Being a Cauchy sequence in (R_i) -metric, there exists $N_{k(j)} \in \mathbb{N}$ such that $a_n - a_m \in R_{k(j)}$ whenever $n, m \geq N_{k(j)}$. Since $R'_k \supset R_{k(j)}$, we see that whenever $n, m \geq N_{k(j)}$, we have

$$a_n - a_m \in R'_k \implies \|a_n - a_m\| \leq c^{-k}$$

hence (a_n) is a Cauchy sequence in the (R'_i) -metric as well. Similarly, we can prove that the Cauchy sequence in the latter metric is a Cauchy sequence in the former metric. Hence both sequences of ideals give rise to the same completion $\hat{R}$.

ii) Let $(r_i + R_i) \in \tilde{R}$. Then to prove that (r_i) forms a Cauchy sequence in R , it suffices to show that for every k , there exists $N_k \in \mathbb{N}$ such that $r_i - r_j \in R_k$ for every $i, j \geq N_k$. However, since $(r_i + R_i) \in \tilde{R}$, if $i < j$ then we have the natural projection map $\lambda_{ji} : R/R_j \rightarrow R/R_i$ and $(r_i + R_i)_{i \in \mathbb{N}}$ satisfies $r_j + R_i = \lambda_{ji}(r_j + R_j) = r_i + R_i$. This shows that $N_k = k$ works perfectly well. If $i, j > k$ then $r_i + R_k = r_j + R_k = r_k + R_k \implies r_i - r_j \in R_k$. Conversely, given a Cauchy sequence (a_n) inside R , we can construct an element of $\tilde{R}$ as follows. For each $k \in \mathbb{N}$, there exists smallest $N_k \in \mathbb{N}$ such that the sequence of cosets $(a_n + R_k)_{n=1}^\infty$ stabilizes so the coset $a_{N_k} + R_k$. This follows from the Cauchy criteria. Then, if $k < \ell$ then we must have $N_k \leq N_\ell$. Notice that if $k < \ell$ then

$$\lambda_{\ell k}(a_{N_\ell} + R_\ell) = a_{N_\ell} + R_k = a_{N_k} + R_k$$

where the last equality holds because $N_k \leq N_\ell$. This proves that $(a_{N_i} + R_i) \in \tilde{R}$ as desired. Hence we obtain an isomorphism $\tilde{R} \rightarrow \hat{R}$ that sends $(r_i + R_i)$ to $\lim r_i$.

iii) Firstly, by the previous exercise, $\{x \in \hat{R} : \|x\| \leq c^{-k}\}$ is a closed set. Secondly, by definition of the norm, $x \in R_k \iff \|x\| \leq c^{-k}$ so the given set contains R_k . Moreover, any closed set containing R_k must also contain the given set. Hence, $\overline{R_k} = \{x \in R : \|x\| \leq c^{-k}\}$. Notice that if $i < k$ then $(R_k + R_i)/R_i = \{1\}$ and if $\ell > k$ then $(R_k + R_\ell)/R_\ell = R_k/R_\ell$ so

$$\varprojlim ((R_k + R_i)/R_i) \cong \varprojlim_{\ell \geq k} (R_k/R_\ell)$$

From part ii, we know that the latter ring is isomorphic to $\widehat{R_k}$. A subset of a complete space $(\hat{R})$ is complete if and only if it is closed hence $\widehat{R_k} = \overline{R_k}$. Additionally, we have $\overline{R_k} \cap R = R_k$. And to prove that $\overline{R_k} + R = \hat{R}$ it suffices to show $\overline{R_k} + R \supset \hat{R}$ since the other inclusion is obvious. So let $(r_i + R_i)_{i=1}^\infty \in \hat{R}$. Then, we can write

$$(r_i + R_i)_{i=1}^\infty = (r_1 + R_1, \dots, r_{k-1} + R_{k-1}, r_k + R_k, \dots, r_k + R_n, \dots) \\ + (0 + R_1, \dots, 0 + R_k, (r_{k+1} - r_k) + R_{k+1}, \dots, (r_n - r_k) + R_n, \dots)$$

In this sum, the first element is equal to $r_k \in R$ and the second element is in $\overline{R_k}$, which proves the last claim.

Finally,

$$\frac{R_k}{R_{k+1}} \cong \frac{R_k}{\overline{R_{k+1}} \cap R_k} \cong \frac{\overline{R_{k+1}} + R_k}{\overline{R_{k+1}}} \cong \frac{\widehat{R_k}}{\overline{R_{k+1}}} \cong \frac{\overline{R_k}}{\overline{R_{k+1}}}$$

iv) If (J_α) is cofinal with (R_i) then we can extract a descending chain of ideals (J'_α) from (J_α) . Then by part i, (J'_α) and (R_i) give rise to the same completion $\hat{R}$. Then, by part ii,

$$\varprojlim (R/J_\alpha) \cong \varprojlim (R/J'_\alpha) \cong \hat{R} \cong \tilde{R}$$

where the first isomorphism is true by the fact that (J_α) form an inverse system directed by reverse inclusion. (ie for every α, β there exists γ such that $J_\alpha \cap J_\beta \supset J_\gamma$.) $\square$

Exercise 6.3

i) Let $f : \mathbb{Q}_p^n \rightarrow \mathbb{Q}_p^m$ be a linear map. It suffices to show f is continuous at 0, which follows if f is bounded near zero. Let $\{e_1, \dots, e_n\}$ be the basis of $\mathbb{Q}_p^n$. Then

$$|f(x)| = \left| f\left(\sum a_i e_i\right) \right| \leq \max_i |a_i| \cdot |f(e_i)| = C \cdot \max_i |a_i| = C \cdot |x|$$

so f is continuous at 0. Here, the metric defined on $\mathbb{Q}_p^n$ is given by $|x| = \max_i |x_i|_p$. Now let V be a vector space of dimension n over the field $\mathbb{Q}_p$. Then, we have a linear map $f : V \rightarrow \mathbb{Q}_p^n$ which induces a topology on V , called the p -adic topology. Let W be a subspace of V . To prove W is closed in V , it is enough to prove $\mathbb{Q}_p^m$ is closed in $\mathbb{Q}_p^{n+m}$ for any $n, m \in \mathbb{Z}$. Now, let $e_1, \dots, e_m$ be a basis for $\mathbb{Q}_p^m$ and let $e_{m+1}, \dots, e_n$ be the remaining basis vectors for $\mathbb{Q}_p^{n+m}$. Then, given $x = (x_1, \dots, x_m, y_1, \dots, y_n) \in \mathbb{Q}_p^{n+m} - \mathbb{Q}_p^n$, let $\delta = \frac{1}{2} \cdot (\min_{i=1}^n |y_i|)$. Now, if $|x - y| < \delta$ then

$$|x - y| = \max_{i,j} \{x_i - a_i, y_j - b_j\} < \delta$$

which implies that at least one coefficient of $e_{m+1}, \dots, e_n$ is non-zero hence $y \in \mathbb{Q}_p^{n+m} - \mathbb{Q}_p^n$. So the complement is open, hence $\mathbb{Q}_p^m$ is closed.

ii) Let $\iota : V \rightarrow A$ be the inclusion map and $\{e_1, \dots, e_n\}$ a basis for V and $\{e_1, \dots, e_{n+m}\}$ a basis for A . Then, suppose wlog that $\|e_1\| \leq \dots \leq \|e_{n+m}\|$. Let $\varepsilon > 0$ be given. Pick $\delta = \frac{\varepsilon}{\|e_n\|}$. Since ι is linear, to prove its continuity, it suffices to prove continuity at the origin. So if $x \in V$ such that $x = \sum_{i=1}^n a_i e_i$ satisfies

$$|x| = \max_i |a_i| < \delta$$

then

$$\|x\| = \max_i |a_i| \cdot \|e_i\| \leq \|e_n\| \cdot \max_i |a_i| < \|e_n\| \cdot \delta < \varepsilon$$

so ι is continuous.

7 The Group Algebra

Exercise 7.1

Using the universal property of inverse limits show that there is a natural epimorphism

$$\pi : \mathbb{Z}_p[[G]] \rightarrow \varprojlim_k (\mathbb{Z}/p^k\mathbb{Z})[G/G_k]$$

Prove that π is an isomorphism and deduce that $\mathbb{Z}_p[[G]] \cong \widehat{R}$.

Proof. Taking $\mathbb{Z}_p[[G]]$ to be R as in the solution to exercise 1.3, we have the following maps for $N \leq M$ and $k > \ell$.

$$\begin{aligned} \mathbb{Z}_p[G/N] &\xrightarrow{\beta_{NM}} \mathbb{Z}_p[G/M] \\ \mathbb{Z}_p[G/N] &\xrightarrow{\pi_k} \mathbb{Z}/p^k\mathbb{Z}[G/G_k] \\ \mathbb{Z}_p[G/N] &\xrightarrow{\pi_\ell} \mathbb{Z}/p^\ell\mathbb{Z}[G/G_\ell] \end{aligned}$$

Suppose $G_{k+1} \leq N \leq G_k$ then π_k is the natural projection of G/N onto G/G_k with coefficients $\mathbb{Z}_p$ getting projected onto $\mathbb{Z}/p^k\mathbb{Z}$. It is obvious that $\pi_\ell = \pi_{k\ell} \circ \pi_k$

and similar commutation relations are satisfied for maps coming out of $\mathbb{Z}_p[G/M]$. Hence, we get a natural map $\pi : \mathbb{Z}_p[[G]] \rightarrow \varprojlim_k (\mathbb{Z}/p^k\mathbb{Z})[G/G_k]$.

π is seen to be injective since an element of $\mathbb{Z}_p[[G]]$ is given by $\lambda(g_N N)_{N \trianglelefteq_o G}$ where $\lambda \in \mathbb{Z}_p$ and this element is sent to 0 in the projective limit if and only if $\lambda = 0$. For surjectivity, it suffices to show that an element of the form $(\lambda_k g_k G_k)_{k \in \mathbb{N}}$ is in the image of π , since π is a homomorphism. We can break apart $(\sum_j \lambda_k^j g_k^j \cdot G_k)_{k \in \mathbb{N}}$ into $(\lambda_k^j g_k^j \cdot G_k)_{k \in \mathbb{N}}$ and find a preimage for each such term. To do this, given $(\lambda_k g_k \cdot G_k)_{k \in \mathbb{N}}$, we consider the element $(g_k G_k)_{k \in \mathbb{N}}$ of G . Since G is pro- p , we know that $\varprojlim (G/N) \cong \varprojlim (G/G_k)$ hence there exists an element $(g_N \cdot N)_{N \trianglelefteq_o G}$ that corresponds to $(g_k \cdot G_k)_{k \in \mathbb{N}}$. Finally, let $\lambda = \sum_{n=0}^{\infty} \lambda_n \cdot p^n \in \mathbb{Z}_p$. Then it is easily seen that

$$\pi(\lambda(g_N \cdot N)_{N \trianglelefteq_o G}) = (\lambda_k g_k \cdot G_k)_{k \in \mathbb{N}}$$

so π is an isomorphism.

To prove the last claim, we already know that $\mathbb{Z}_p[[G]] \cong \varprojlim_k (\mathbb{Z}/p^k\mathbb{Z})[G/G_k] \cong \varprojlim_k (R/(I_k + p^k R))$, where

$$I_k = \ker(R \rightarrow \mathbb{Z}_p[G/G_k])$$

and therefore

$$I_k + p^k R = \ker(R \rightarrow (\mathbb{Z}/p^k\mathbb{Z})[G/G_k])$$

Set $J = I_1 + pR$. By Lemma 7.1, we know that J^k and $I_k + p^k R$ are cofinal hence by exercise 6.2, they give rise to the same inverse limit. Then, we obtain

$$\mathbb{Z}_p[[G]] \cong \varprojlim_k (\mathbb{Z}/p^k\mathbb{Z})[G/G_k] \cong \varprojlim_k (R/(I_k + p^k R)) \cong \varprojlim_k R/J^k \cong \widehat{R}$$

where the last congruence follows from exercise 6.2 part ii). $\square$

Exercise 7.2

Let $x \in G$. Show that if $x \in G_k$, then $\|x - 1\| \leq p^{-k}$ and that if, $x \in I_k + pR$ then $x \in G_k$. Deduce that for each k , there exists $m > 1$ such that

$$G \cap \overline{B}(1, p^{-m}) \subseteq G_k \subseteq G \cap \overline{B}(1, p^{-k})$$

Show that if G is uniform then $G_k = G \cap \overline{B}(1, p^{-k})$.

Proof. Let $x \in G_k$. Notice that $\|x - 1\| \leq p^{-k} \iff x - 1 \in J^m \setminus J^{m+1}$ for some m satisfying $m \geq k$ since then $\|x - 1\| = p^{-m} \leq p^{-k}$. Hence, it suffices to show that $x - 1 \in J^k$. By lemma 7.1, $J^k \supseteq I_k + p^k R$ and $x - 1 \in G_k - 1 \implies x - 1 \in I_k$ so $x - 1 \in J^k$ as desired. Now suppose $x \in I_k + pR$. Then, since $x \in G$, we must have $x = (x - 1) + 1$ as a decomposition into I_k and R parts, which means

$x - 1 \in I_k \implies x \in G_k$. Then for fixed k , by Lemma 7.1, there exists m , in fact $m = |G/G_k|$, such that

$$I_k + pR \supseteq J^m$$

then we have

$$G_k \supseteq G \cap I_k + pR$$

so

$$G \cap \overline{B}(1, p^{-m}) \subset G_k \subset G \cap \overline{B}(1, p^{-k})$$

as desired. Therefore, the norm topology on R induced the original topology on G as a subspace of R .

Now suppose G is uniform. Then, let $x \in G \cap \overline{B}(1, p^{-k})$. Then,

$$x = \sum_{\alpha \in \mathbb{N}^d} \lambda_\alpha \mathbf{b}^\alpha$$

where $b_i = a_i - 1$ and $\lambda_\alpha \in \mathbb{Z}_p$. Then, we also have $x = a_1^{\lambda_1} \dots a_d^{\lambda_d}$ for some $\lambda_i \in \mathbb{Z}_p$ as well. By exercise 3, we can write

$$\sum_{\alpha \in \mathbb{N}^d} \lambda_\alpha \mathbf{b}^\alpha = \sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \mathbf{b}^\alpha$$

hence

$$\lambda_\alpha = \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d}$$

Since $\|x - 1\| < p^{-k}$, by theorem 7.5

$$\|x - 1\| = \sup_{\alpha \in \mathbb{N}^d} p^{-\langle \alpha \rangle} |\lambda_\alpha|$$

where if $\alpha = 0$ then $\lambda_\alpha = \lambda_\alpha - 1$. Take $\alpha = \epsilon_i$ then

$$p^{-1} |\lambda_{\epsilon_i}| = p^{-1} |\lambda_i| \leq p^{-k} \implies |\lambda_i| \leq p^{-(k-1)} \implies p^{k-1} |\lambda_i|$$

and so $x \in G_k$. □

Exercise 7.3

i) Let $a_n \rightarrow \lambda$ as $n \rightarrow \infty$ and $a_n \in \mathbb{Z}$. Such a sequence exists since $\mathbb{Z}$ is dense in the completion $\mathbb{Z}_p$. Then,

$$\binom{a_n}{k} = \frac{a_n(a_n - 1) \dots (a_n - r + 1)}{r!} = \frac{P(a_n)}{r!}$$

where $P(x) := x(x - 1) \dots (x - r + 1)$. Then, $\binom{a_n}{k} \in \mathbb{Z}$ and since $P(x)$ is continuous $P(a_n) \xrightarrow{n \rightarrow \infty} \binom{\lambda}{r}$ hence $\binom{\lambda}{r} \in \mathbb{Z}_p$.

ii) Let $\varepsilon > 0$ be given. We want to show that

$$\left\| \mathbf{a}^\lambda - \sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \right\| < \varepsilon$$

To do this we will use non-archimedian metric property and show that

$$\begin{aligned} \left\| \mathbf{a}^\lambda - \sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \right\| &= \left\| (\mathbf{a}^\lambda - \mathbf{a}^\beta) + \left[\mathbf{a}^\beta - \sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \right] \right\| \\ &< \max \left\{ \|(\mathbf{a}^\lambda - \mathbf{a}^\beta)\|, \left\| \mathbf{a}^\beta - \sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \right\| \right\} \\ &< \varepsilon \end{aligned}$$

where $\beta \in \mathbb{N}^d$ is chosen in a certain way. Note that by lemma 7.8 we already have

$$\mathbf{a}^\beta = \sum_{\alpha \in \mathbb{N}^d} \binom{\beta_1}{\alpha_1} \dots \binom{\beta_d}{\alpha_d} \mathbf{b}^\alpha$$

Now, we need to show that

$$\|(\mathbf{a}^\lambda - \mathbf{a}^\beta)\| < \varepsilon \quad (3)$$

$$\left\| \mathbf{a}^\beta - \sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \right\| < \varepsilon \quad (4)$$

for a suitable choice of β . Choose $N \in \mathbb{N}$ such that $p^{-N} < \varepsilon$ and pick β_i for $i = 1, \dots, d$ such that

$$\lambda_i \equiv \beta_i \pmod{p^N}$$

Then,

$$a_i^{\lambda_i} - a_i^{\beta_i} = a_i^{\beta_i} (a_i^{\lambda_i - \beta_i} - 1) \in \mathbb{Z}_p[G](G_{N+1} - 1) = I_{N+1} \implies \|a_i^{\lambda_i} - a_i^{\beta_i}\| \leq p^{-(N+1)}$$

so

$$\begin{aligned} \|\mathbf{a}^\lambda - \mathbf{a}^\beta\| &= \|a_1^{\lambda_1} \dots a_r^{\lambda_r} - a_1^{\beta_1} a_2^{\lambda_2} \dots a_r^{\lambda_r} \\ &\quad + a_1^{\beta_1} a_2^{\lambda_2} \dots a_r^{\lambda_r} - a_1^{\beta_1} a_2^{\beta_2} a_3^{\lambda_3} \dots a_r^{\lambda_r} \\ &\quad + \dots + u_1^{\beta_1} \dots u_{r-1}^{\beta_{r-1}} u_r^{\lambda_r} - u_1^{\beta_1} \dots u_r^{\beta_r}\| \\ &\leq \max_i \|u_i^{\lambda_i} - u_i^{\beta_i}\| \cdot \max_i \|u_i^{\lambda_i + \beta_i}\|^{r-1} \\ &\leq p^{-(N+1)} \cdot 1 \\ &< \varepsilon \end{aligned}$$

This proves that (3) is satisfied. Now we need to show (4) which is the same thing as showing

$$\left\| \sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \mathbf{b}^\alpha - \sum_{\alpha \in \mathbb{N}^d} \binom{\beta_1}{\alpha_1} \dots \binom{\beta_d}{\alpha_d} \mathbf{b}^\alpha \right\| < \varepsilon$$

Notice that by the non-archimedian property,

$$\begin{aligned} \left\| \sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \mathbf{b}^\alpha - \sum_{\alpha \in \mathbb{N}^d} \binom{\beta_1}{\alpha_1} \dots \binom{\beta_d}{\alpha_d} \mathbf{b}^\alpha \right\| &\leq \max_{\alpha \in \mathbb{N}^d} \left\| \left(\binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} - \binom{\beta_1}{\alpha_1} \dots \binom{\beta_d}{\alpha_d} \right) \mathbf{b}^\alpha \right\| \\ &\leq \max_{\alpha \in \mathbb{N}^d} \left\{ \|\mathbf{b}^\alpha\|, \left| \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} - \binom{\beta_1}{\alpha_1} \dots \binom{\beta_d}{\alpha_d} \right| \right\} \end{aligned}$$

Now, for a given $\alpha \in \mathbb{N}^d$ if $\langle \alpha \rangle > N$ then

$$\|\mathbf{b}^\alpha\| \leq \prod_{i=1}^d \|b_i^{\alpha_i}\| \leq p^{-\langle \alpha \rangle} \leq p^{-N} < \varepsilon$$

and if $\langle \alpha \rangle < N$ then we have

$$\begin{aligned} \left| \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} - \binom{\beta_1}{\alpha_1} \dots \binom{\beta_d}{\alpha_d} \right| &\leq \max_i \left| \binom{\lambda_i}{\alpha_i} - \binom{\beta_i}{\alpha_i} \right| \\ &\leq \left(\max_i |\lambda_i - \beta_i| \right) \cdot \prod_{j=1}^d \frac{1}{|\alpha_j!|} \\ &\leq p^{-N} \frac{1}{|N!|} < \varepsilon \end{aligned}$$

where the last inequality follows from the combinatorics fact that

$$\prod_j \alpha_j! \mid \langle \alpha \rangle! \mid N!$$

and the property of the p-adic absolute value implies

$$\frac{1}{\prod_j |\alpha_j!|} < \frac{1}{|\langle \alpha \rangle!|} < \frac{1}{N!}$$

so (4) is satisfied as well and we are done.

iii) Then an element of $R = \mathbb{Z}_p[G]$ is a finite sum

$$\sum_{g \in G} a_g \cdot g$$

where $a_g \in \mathbb{Z}_p$. Since G is uniform, $G = \overline{\langle a_1 \rangle} \dots \overline{\langle a_d \rangle}$ so each g in the above sum can be expressed as $g = a_1^{\lambda_1} \dots a_d^{\lambda_d}$ with $\lambda_i \in \mathbb{Z}_p$. By part ii), each such product is given a sum of the form $\sum_{\alpha \in \mathbb{N}^d} \binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} \mathbf{b}^\alpha$. Addind them all in the main sum $\sum a_g g$, we see that each element of R can be expressed as a sum of the form

$$\sum_{\alpha \in \mathbb{N}^d} \lambda_\alpha \mathbf{b}^\alpha \quad \lambda_\alpha \in \mathbb{Z}_p$$

as desired.

iv) As the hint suggests, let (w_i) be a Cauchy sequence in A . By part ii),

$$w_i = \sum_{\alpha \in \mathbb{N}^d} \lambda_{\alpha i} \mathbf{b}^\alpha$$

where $\lambda_{\alpha i} \in \mathbb{Q}_p$. Then given $\varepsilon > 0$, for large enough i, j we have

$$\begin{aligned} \|w_i - w_j\| &= \left\| \sum_{\alpha \in \mathbb{N}^d} (\lambda_{\alpha i} - \lambda_{\alpha j}) \mathbf{b}^\alpha \right\| \\ &= \sup_{\alpha} p^{-\langle \alpha \rangle} |\lambda_{\alpha i} - \lambda_{\alpha j}| \\ &< \varepsilon \end{aligned}$$

Fix $\alpha \in \mathbb{N}^d$. Then for large enough i, j

$$\begin{aligned} p^{-\langle \alpha \rangle} |\lambda_{\alpha i} - \lambda_{\alpha j}| &\leq \sup_{\beta \in \mathbb{N}^d} p^{-\langle \beta \rangle} |\lambda_{\beta i} - \lambda_{\beta j}| \\ &< \frac{\varepsilon}{p^{\langle \alpha \rangle}} \end{aligned}$$

Hence

$$|\lambda_{\alpha i} - \lambda_{\alpha j}| < \varepsilon$$

Then for fixed α , $\lambda_{\alpha i}$ converges to an element $\lambda_\alpha \in \mathbb{Q}_p$. Given $\varepsilon > 0$, there exists w_i such that $\|w - w_i\| < \varepsilon$. Moreover,

$$\begin{aligned} \left\| \sum_{\alpha \in \mathbb{N}^d} \lambda_\alpha \mathbf{b}^\alpha - w_i \right\| &= \left\| \sum_{\alpha \in \mathbb{N}^d} (\lambda_\alpha - \lambda_{\alpha i}) \mathbf{b}^\alpha \right\| \\ &= \sup_{\alpha \in \mathbb{N}^d} p^{-\langle \alpha \rangle} |\lambda_\alpha - \lambda_{\alpha i}| \\ &< \varepsilon \end{aligned}$$

for large enough i as well. Then,

$$\left\| w - \sum_{\alpha} \lambda_\alpha \mathbf{b}^\alpha \right\| \leq \max(\|w - w_i\|, \|w_i - \sum_{\alpha} \lambda_\alpha \mathbf{b}^\alpha\|) < \varepsilon$$

Hence $w = \sum_{\alpha \in \mathbb{N}^d} \lambda_\alpha \mathbf{b}^\alpha$ as desired.

Exercise 7.8

Notice that

$$pM_n(\mathbb{Z}_p) = \{x \in M_n(\mathbb{Q}_p) : \|x\| \leq p^{-1}\}$$

Hence

$$\exp : pM_n(\mathbb{Z}_p) \rightarrow I + pM_n(\mathbb{Z}_p)$$

By proposition 6.22,

$$\exp(pA) = \sum_{n=0}^{\infty} \frac{1}{n!} (pA)^n = I + p \sum_{n=1}^{\infty} \frac{1}{n!} p^{n-1} A^n \in \Gamma$$

and for $I + pA \in \Gamma$, we have

$$\log(I + pA) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (pA)^n = (pA) \cdot \left(\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (pA)^{n-1} \right) \in pM_n(\mathbb{Z}_p)$$

8 p-adic Analytic Groups

Exercise 8.1

Let X, Y be p-adic analytic manifolds. Prove that the projection map $X \times Y \rightarrow Y$ is an analytic function.

Proof. Let $(x, y) \in X \times Y$. Then there are charts (U, φ) containing x and (V, ψ) containing y . Then, it suffices to show that the map $\varphi \circ f \circ (\varphi \times \psi)^{-1} : \mathbb{Z}_p^n \times \mathbb{Z}_p^m \rightarrow \mathbb{Z}_p^n$ is analytic. Let $y = (y_1, \dots, y_{n+m}) \in \mathbb{Z}_p^{n+m}$ and write $g = \varphi \circ f \circ (\varphi \times \psi)^{-1}$. Then $g(y) = (y_1, \dots, y_n) \in \mathbb{Z}_p^n$. Let $h = 0$ then for $z \in B(y, 1)$, we can write $z = y + x$ for some $x \in \mathbb{Z}_p$. Then, if we write $g = (g_1, \dots, g_n)$, we have

$$g_i(y + x) = F_i(x)$$

where $F_i \in \mathbb{Q}_p[X]$ is given by

$$F_i(X) = \sum_{w \in W(X)} a_w w$$

where

$$a_w = \begin{cases} y_i & w = \emptyset \\ 1 & w = x_i \\ 0 & \text{otherwise} \end{cases}$$

This shows that g is analytic at y . Hence projection map is an analytic map. $\square$

Exercise 8.2

Proof. Suppose f is analytic. $\square$

Exercise 8.3

i) This is obvious since $\exp : \Lambda \rightarrow G$ is a Lie algebra isomorphism and it is a homeomorphism.

ii) We have the following functions.

$$\begin{aligned}\mathcal{L}(X) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} X^n \\ \mathcal{E}(X) &= \sum_{n=0}^{\infty} \frac{1}{n!} X^n \\ H(X_1, \dots, X_d) &= \mathcal{E}(X_1) \dots \mathcal{E}(X_d) - 1\end{aligned}$$

By theorem 6.24, it suffices to prove that

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{n} \right| \tau^n = 0$$

where

$$\tau = \sup \left\{ \left\| \frac{y_1^n}{(n_1)!} \right\| \dots \left\| \frac{y_d^n}{(n_d)!} \right\| : (n_1, \dots, n_d) \in \mathbb{N}^d - (0, \dots, 0) \right\}$$

Then we can write

$$\Phi^{(d)}(y_1, \dots, y_d) = (\mathcal{L} \circ \mathcal{H})(y_1, \dots, y_d) = \mathcal{L}(\mathcal{H}(y_1, \dots, y_d)) = \log(\mathcal{E}(y_1) \dots \mathcal{E}(y_d)) = \log(\exp(y_1) \dots \exp(y_d))$$

Now, by equation (5) in chapter 6

$$\tau \leq \|y_1\| \dots \|y_d\| \leq p^{-d}$$

so

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{n} \right| \tau^n = 0$$

as desired.

iii) Notice that $\lambda_i \in \mathbb{Z}_p$ for each $i = 1, \dots, d$. Hence $\|\lambda_i x_i\| = |\lambda_i| \cdot \|x_i\| \leq p^{-1} \implies \lambda_i x_i \in \widehat{A_0}$. Then by part ii)

$$\begin{aligned}\Phi^{(d)}(\lambda_1 x_1, \dots, \lambda_d x_d) &= \log(\exp(\lambda_1 x_1) \dots \exp(\lambda_d x_d)) \\ &= \log(\exp(x_1)^{\lambda_1} \dots \exp(x_d)^{\lambda_d}) \\ &= \log(a_1^{\lambda_1} \dots a_d^{\lambda_d}) \\ &= \mu_1 x_1 + \dots + \mu_d x_d\end{aligned}$$

where $\psi \circ \phi^{-1}(\lambda_1, \dots, \lambda_d) = (\mu_1, \dots, \mu_d)$. If we let $\eta : \Lambda \rightarrow \mathbb{Z}_p^d$ be the chart given by

$$\eta(\nu_1 x_1 + \dots + \nu_d x_d) = (\nu_1, \dots, \nu_d)$$

then

$$\psi \circ \phi^{-1}(\lambda_1, \dots, \lambda_d) = \nu \circ \log \circ \phi^{-1}(\lambda_1, \dots, \lambda_d) = \nu \circ \Phi^{(d)}(\lambda_1, \dots, \lambda_d)$$

the last function is analytic, hence so is $\psi \circ \phi^{-1}$.

iv) Well, we compute the coefficient of each term. If $\langle \beta \rangle = n$ then the coefficient of $x_1^{\beta_1} \dots x_d^{\beta_d}$ appearing in $\frac{1}{n!}(\mu_1 x_1 + \dots + \mu_d x_d)^n$ is

$$\frac{1}{n!} \frac{n!}{\beta_1! \dots \beta_d!} \mu_1^{\beta_1} \dots \mu_d^{\beta_d} = \frac{\mu_1^{\beta_1} \dots \mu_d^{\beta_d}}{\beta_1! \dots \beta_d!}$$

so

$$y_\beta = \frac{1}{\beta_1! \dots \beta_d!} \cdot x_1^{\beta_1} \dots x_d^{\beta_d} \in \hat{A}$$

and we have

$$\frac{1}{n!}(\mu_1 x_1 + \dots + \mu_d x_d)^n = \sum_{\langle \beta \rangle = n} \mu_1^{\beta_1} \dots \mu_d^{\beta_d} \cdot y_\beta$$

and by exercise 7.3 part iv), each $y_\beta \in \hat{A}$ can be expressed as

$$y_\beta = \sum_{\alpha \in \mathbb{N}^d} d_{\alpha\beta} \mathbf{b}^\alpha$$

where $d_{\alpha\beta} \in \mathbb{Q}_p$ for every α, β . Now, if $\psi(x) = (\mu_1, \dots, \mu_d)$ then $\psi^{-1}(\mu_1, \dots, \mu_d) = x$ and $\psi^{-1} = \exp \circ \eta^{-1}$ so $\psi^{-1}(\mu_1, \dots, \mu_d) = \exp(\mu_1 x_1 + \dots + \mu_d x_d)$ hence we have

$$\begin{aligned} x = \exp(\mu_1 x_1 + \dots + \mu_d x_d) &= \sum_{n \geq 0} \frac{1}{n!} (\mu_1 x_1 + \dots + \mu_d x_d)^n \\ &= \sum_{n \geq 0} \sum_{\langle \beta \rangle = n} \mu_1^{\beta_1} \dots \mu_d^{\beta_d} \sum_{\alpha \in \mathbb{N}^d} d_{\alpha\beta} \mathbf{b}^\alpha \\ &= \sum_{\alpha \in \mathbb{N}^d} \left(\sum_{\beta \in \mathbb{N}^d} \mu_1^{\beta_1} \dots \mu_d^{\beta_d} \right) \cdot \mathbf{b}^\alpha \end{aligned}$$

On the other hand, if $\phi(x) = (\lambda_1, \dots, \lambda_d)$ then $x = a_1^{\lambda_1} \dots a_d^{\lambda_d}$ and so

$$\sum_{\alpha \in \mathbb{N}^d} \binom{\mu_1}{\alpha_1} \dots \binom{\mu_d}{\alpha_d} \mathbf{b}^\alpha = x = \sum_{\alpha \in \mathbb{N}^d} \left(\sum_{\beta \in \mathbb{N}^d} \mu_1^{\beta_1} \dots \mu_d^{\beta_d} \right) \cdot \mathbf{b}^\alpha$$

which shows

$$\binom{\lambda_1}{\alpha_1} \dots \binom{\lambda_d}{\alpha_d} = \sum_{\beta \in \mathbb{N}^d} d_{\alpha\beta} \mu_1^{\beta_1} \dots \mu_d^{\beta_d}$$

Then to see that $\phi \circ \psi^{-1}$ is analytic, notice that

$$\phi \circ \psi^{-1}(\mu_1, \dots, \mu_d) = \phi(x) = (\lambda_1, \dots, \lambda_d)$$

and by taking $\alpha = \epsilon_i$ for $i = 1, \dots, d$ we see that

$$\lambda_i = \sum_{\beta \in \mathbb{N}^d} d_{\epsilon_i \beta} \mu_1^{\beta_1} \dots \mu_d^{\beta_d}$$

so each coordinate function $(\phi \circ \psi^{-1})_i(\mu_1, \dots, \mu_d)$ is represented by an analytic function. Hence $\phi \circ \psi^{-1}$ is analytic.

v) Since $(G, \phi, d) \sim (G, \psi, d)$ it suffices to prove $(G, \theta, d) \sim (G, \phi, d)$ and now if we write

$$\mu_i = \sum_{j=1}^d a_{ji} z_j$$

then

$$\theta \circ \phi^{-1}(\mu_1, \dots, \mu_d) = (t_1, \dots, t_d)$$

where

$$t_i = \mu_1 a_{i1} + \mu_2 a_{i2} + \dots + \mu_d a_{id}$$

which shows this map is analytic. By symmetry, $\phi \circ \theta^{-1}$ is also analytic.

Exercise 8.7

Let X be transversal to the right cosets of $G \cap GL_n(\mathbb{Z}_p)$. Then we know that automatically,

$$G \subset GL_n(\mathbb{Z}_p) \cdot X$$

Now, to see that $GL_n(\mathbb{Z}_p) \cap G$ is open in G we need to show that $GL_n(\mathbb{Z}_p)$ is open in $GL_n(\mathbb{Q}_p)$.

$$GL_n(\mathbb{Z}_p) = GL_n(\mathbb{Q}_p) \cap M_n(\mathbb{Z}_p)$$

and the topology on $GL_n(\mathbb{Q}_p)$ is the subspace topology inherited from $M_n(\mathbb{Q}_p)$. So it suffices to show $M_n(\mathbb{Z}_p)$ is open in $M_n(\mathbb{Q}_p)$. To prove the latter claim, it suffices to show that there exists an open neighborhood around $I \in GL_n(\mathbb{Z}_p)$ contained inside $GL_n(\mathbb{Z}_p)$. However it is easy too see that $B_\delta(I) \subset GL_n(\mathbb{Z}_p)$ whenever $\delta \leq 1$. Since G is compact and $G \cap GL_n(\mathbb{Z}_p)$ is open in G , X is finite.

Now, X acts on $\mathbb{Z}_p^n$ by

$$A(\mathbb{Z}_p^n) = \sum_{i=1}^n \mathbb{Z}_p \cdot A(e_i)$$

where $A \in X$ and e_i is the standard basis element of $\mathbb{Z}_p^n$. Notice that if $A \in GL_n(\mathbb{Z}_p)$ then $A(\mathbb{Z}_p^n) = \mathbb{Z}_p^n$. So, if we write $G' = G \cap GL_n(\mathbb{Z}_p)$ we have

$$G = \bigcup_{x \in X} xG'$$

and if $g \in G$ we can write $g = xh$ for $x \in X, h \in G'$ and

$$g(\mathbb{Z}_p^n) = xh(\mathbb{Z}_p^n) = x(\mathbb{Z}_p^n)$$

so

$$\{g(\mathbb{Z}_p^n) : g \in G\} = \{x(\mathbb{Z}_p^n) : x \in X\}$$

Then consider the lattice of $\mathbb{Q}_p^n$ given by

$$M = \sum_{x \in X} x(\mathbb{Z}_p^n)$$

Notice that M is fixed by G since if $gx = \sigma_x h_x \in G$ where $\sigma_x \in X, h_x \in G'$ then

$$g(M) = \sum_{x \in X} gx(\mathbb{Z}_p^n) = \sum_{x \in X} \sigma_x h_x(\mathbb{Z}_p^n) = \sum_{x \in X} \sigma_x(\mathbb{Z}_p^n) = M$$

So

$$G \subset K_M$$

where K_M is the stabilizer subgroup inside $GL_n(\mathbb{Q}_p)$ of the lattice M , ie

$$K_M := \{g \in GL_n(\mathbb{Q}_p) : g(M) = M\}$$

All lattices of $\mathbb{Q}_p^n$ are given by $A(\mathbb{Z}_p^n)$ where $A \in GL_n(\mathbb{Q}_p)$. Then, $M = y(\mathbb{Z}_p^n)$ for some $y \in GL_n(\mathbb{Q}_p)$. Then

$$\begin{aligned} G \leq K_M &= \{h \in GL_n(\mathbb{Q}_p) : h(M) = M\} \\ &= \{h \in GL_n(\mathbb{Q}_p) : h(y(\mathbb{Z}_p^n)) = y(\mathbb{Z}_p^n)\} \\ &= \{h \in GL_n(\mathbb{Q}_p) : y^{-1}hy(\mathbb{Z}_p^n) = \mathbb{Z}_p^n\} \\ &= \{h \in GL_n(\mathbb{Q}_p) : y^{-1}hy \in GL_n(\mathbb{Z}_p)\} \\ &= yGL_n(\mathbb{Z}_p)y^{-1} \end{aligned}$$

This prove that every compact subgroup of $GL_n(\mathbb{Q}_p)$ is contained in a conjugate of $GL_n(\mathbb{Z}_p)$.

9 Lie Theory

Exercise 9.1

The recurrence relation gives

$$g(m+1) - g(m) = f(m)$$

then write

$$f(T) = \sum_{k=1}^n a_k \binom{T}{k}$$

for $a_k \in \mathbb{Q}_p$ and set

$$g(T) = \sum_{k=1}^n a_k \binom{T}{k+1}$$

Exercise 9.2

Suppose $a_k \in \mathbb{Z}_p$ for $k \leq j \leq d$ then

$$g(j) = a_0 + j \cdot a_1 + \cdots + j \cdot a_{j-1} + a_j \in \mathbb{Z}_p \implies a_j \in \mathbb{Z}_p$$

Exercise 9.3

By corollary 9.13, H^p is a standard group, which is a p-adic analytic group, so by proposition 8.15, it suffices to find, for each $g \in G$, an open neighbourhood V_g of the identity in H^p such that $gV_gg^{-1} \subset H^p$ and $k_g : V_g \rightarrow H^p$ given by $x \mapsto gxg^{-1}$ is analytic. Then let m be large enough to put $V_g = H^{p^m} = P_{m+1}(H) \subset H^p \cap g^{-1}H^pg$. Then the induced atlas on V_g is given by $(V_g, \psi|_{V_g}, d)$ and $\psi : V_g \rightarrow p^m\mathbb{Z}_p$. Since left and right multiplication is a continuous map, and $V_g = P_{m+1}(H)$ is also a standard group with respect to the induced manifold structure, $k_g : V_g \rightarrow H^p$ is a continuous map between p-adic analytic groups. Hence it is analytic by Theorem 9.4. Thus, the p-adic analytic manifold structure on H extends to give a p-adic analytic manifold structure on G .

Exercise 9.4

i) By Ado's theorem, there exists a faithful representation

$$\phi : L \rightarrow M_n(\mathbb{Q}_p)$$

Since L is a finitely-generated $\mathbb{Z}_p$ module, there exists $i \in \mathbb{N}$ such that

$$p^i\phi(L) \subset M_n(\mathbb{Z}_p)$$

Then,

$$\phi(p^{i+1}L) \subset pM_n(\mathbb{Z}_p)$$

ii) L is viewed as a subalgebra of $M_n(\mathbb{Q}_p)$, which is a normed $\mathbb{Q}_p$ -algebra and L_0 , being a subalgebra of $pM_n(\mathbb{Z}_p)$ satisfies

$$\forall x \in L_0, \quad \|x\| \leq p^{-1}$$

so

$$\exp : pM_n(\mathbb{Z}_p) \rightarrow 1 + pM_n(\mathbb{Z}_p)$$

is well-defined. Moreover, L_0 is a powerful Lie algebra since if $x, y \in L_0$ we can associate x, y with pA, pB where $A, B \in M_n(\mathbb{Z}_p)$. Then,

$$[pA, pB] = p \cdot [A, pB] = p \cdot \phi(z)$$

for some $z \in L_0$ since $\phi(L_0)$ is a Lie subalgebra of $pM_n(\mathbb{Z}_p)$ so $[A, pB] \in \phi(L_0)$. This proves L_0 is powerful so $x*y = \Phi(x, y)$ is well-defined. Then, by proposition 6.27,

$$\exp(x * y) = \exp(\Phi(x, y)) = \exp(\log(\exp x \cdot \exp y)) = \exp x \cdot \exp y$$

The last equality from corollary 6.25. Identifying L_0 with a Lie subalgebra of $pM_n(\mathbb{Z}_p)$, the above proves that $(L_0, *)$ is indeed a group.

iii) Now,

$$\log : 1 + pM_n(\mathbb{Z}_p) \rightarrow pM_n(\mathbb{Z}_p)$$

is strictly analytic, and maps $\exp L_0$ to L_0 . By proposition 6.19, $\log$ is continuous then $\log^{-1}(L_0) = \exp L_0$ is closed in $1 + pM_n(\mathbb{Z}_p)$ since L_0 is closed in $pM_n(\mathbb{Z}_p)$. $\log$ is both a group homomorphism and a homeomorphism when restricted to $\exp L_0$ onto L_0 . By theorem 5.2, $1 + pM_n(\mathbb{Z}_p) = \Gamma_1$ is a uniform pro-p group and by example 8.19 (iv) it is a p-adic analytic group. Then by theorem 6.4, $\exp L_0$ is a p-adic analytic group. If (U, ψ, d) is a chart of $\exp L_0$ then $(\log U, \psi \circ \exp, d)$ is a chart of L_0 so it is a p-adic analytic group. Moreover, since closed subgroups of pro-p groups are pro-p, $\exp L_0$ is pro-p and since $\log$ is a group homomorphism and is a homeomorphism, L_0 is also pro-p. If L_0 isn't torsion-free then there exists $x \in L_0$ such that $x^k = 1$ for some $k \in \mathbb{N}$. Then $(\exp x)^k = 1$ but $\exp x \in 1 + pM_n(\mathbb{Z}_p)$ which is uniform, so is torsion-free. This is a contradiction hence L_0 is torsion-free.

Exercise 9.5

Exercise 9.6

Let $H_1 \leq_o A, H_2 \leq_o B$ be uniform pro-p groups such that $f(H_1) \leq H_2$. Then,

$$q \otimes h \in \ker f^* \iff q \otimes f(h) = 0 \iff f(h) = 0 \iff h \in \ker f|_{L_{H_1}} \implies \ker f^* = \mathbb{Q} \otimes \ker f|_{L_{H_1}}$$

Notice that $f|_{L_{H_1}} = H_1 \cap N$. So we need to show $H_1 \cap N$ is a uniform open subgroup of N . It is open in N since H_1 is open in A . Moreover, f is continuous so $N = \ker f = f^{-1}(0)$ is closed in A so $H_1 \cap N \trianglelefteq_c H_1$. If $H_1/H_1 \cap N$ is uniform, then $H_1 \cap N$ is uniform by Theorem 4.31 (ii). $H_1/H_1 \cap N$ is pro-p since H_1 is pro-p and $H_1 \cap N$ is closed in H_1 . Moreover, $H_1/H_1 \cap N$ is powerful by Lemma 2.2 (ii) then it suffices to prove that $H_1/H_1 \cap N$ is torsion-free. Suppose $x \in H_1 - (H_1 \cap N)$ and $x^n \in H_1 \cap N$ for some $n \in \mathbb{N}$. Then, $f(x^n) = f(x)^n = 0$ and $f(x) \in f(H_1) \leq H_2$ implies that $f(x)$ is a torsion element in H_2 , so $f(x) = 0$. Hence $H_1/H_1 \cap N$ is torsion free and so $H_1 \cap N$ is a uniform open subgroup of N . $\ker f^* = \mathcal{L}(N)$.

Exercise 9.7

Let $H_1 \leq_o A, H_2 \leq_o B$ be uniform subgroups such that $f(H_1) \leq H_2$. Then,

$$\begin{aligned} \dim(\text{Im } f^*) &= \dim \mathcal{L}(A) - \dim \mathcal{L}(N) \\ &= \dim H_1 - \dim(\ker f|_{H_1}) \end{aligned}$$

Now, consider the short exact sequence

$$1 \rightarrow \ker f|_{H_1} \xrightarrow{\iota} H_1 \xrightarrow{f} H_2 \rightarrow 1$$

Then $\iota : \ker f|_{H_1} \rightarrow H_1$ is the inclusion map, which is an open map, hence by exercise 8.5 we have

$$\dim H_1 = \dim H_2 + \dim(\ker f|_{H_1})$$

so

$$\dim(\operatorname{Im} f^*) = \dim H_1 - \dim(\ker f|_{H_1}) = \dim H_2 = \dim \mathcal{L}(B)$$

hence f^* is surjective.

Exercise 9.8

Exercise 9.9

i) $G = GL_n(\mathbb{Q}_p)$. Recall that

$$\Gamma = \{g \in GL_n(\mathbb{Z}_p) : g \equiv 1 \pmod{p}\}$$

is an open uniform pro- p subgroup of G . And by exercise 7.8 part ii, $\log \Gamma = pM_n(\mathbb{Z}_p)$ and since $L_\Gamma \cong \log \Gamma$, we can write

$$\mathcal{L}(G) = \mathbb{Q}_p \otimes L_\Gamma = \mathbb{Q}_p \otimes pM_n(\mathbb{Z}_p) = \mathfrak{gl}_n(\mathbb{Q}_p)$$

ii) For $\mu \in p\mathbb{Z}_p$

$$\begin{aligned} \log(1 + \mu e_{ij}) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (\mu e_{ij})^n \\ &= \begin{cases} \log(1 + \mu) e_{ii} & i = j \\ \mu e_{ij} & i \neq j \end{cases} \end{aligned}$$

Hence

$$\operatorname{tr}(\log g) = \log(\det g)$$

$\det^* : M_n(\mathbb{Q}_p) \rightarrow \mathbb{Q}_p$ is given by $1 \otimes \det|_{L_\Gamma \cong \log \Gamma}$ and the above shows

$$\det g = \exp \circ \operatorname{tr} \circ \log(g)$$

for all $g \in L_\Gamma$ so

$$(\det)^* = \operatorname{tr}$$

iii) Notice that $\ker \det = SL_n(\mathbb{Q}_p)$ and $\ker \operatorname{tr} = \mathfrak{sl}_n(\mathbb{Q}_p)$ and by exercise 6,

$$\mathcal{L}(SL_n(\mathbb{Q}_p)) = \mathcal{L}(\ker \det) = \ker(\det)^* = \ker \operatorname{tr} = \mathfrak{sl}_n(\mathbb{Q}_p)$$

Exercise 9.10

i) $Ad_\Lambda : G \rightarrow GL_n(\Lambda)$ is given by $Ad_\Lambda(g)(u) = g^{-1}ug$. Notice if $u = \log h$ then

$$g^{-1}(\log h)g = \log(g^{-1}hg) \in \Lambda$$

Moreover,

$$(Ad_\Lambda(g) - 1)(\log h) = \log(g^{-1}hg) - \log h \dots$$

ii) $g \in G, v = \log g \in \Lambda$ by exercise 6.12

$$\log(Ad_\Lambda(g)) = ad_\Lambda(v)$$

iii)

$$\begin{aligned} Ad(g)(h) - h &= ghg^{-1} + (-h) \\ &= \lim_{n \rightarrow \infty} ((ghg^{-1})^{p^n} h^{-p^n})^{p^{-n}} \\ &= \lim_{n \rightarrow \infty} (gh^{p^n} g^{-1} h^{-p^n})^{p^{-n}} \\ &= \lim_{n \rightarrow \infty} [g^{-1}, h^{-p^n}]^{p^{-n}} \in G_2 = pG \end{aligned}$$

so $Ad(g) \in \Gamma$.

Let $T \in GL(\Lambda)$. We will show

$$\log \circ (\exp \circ T \circ \log) = \exp \circ \log(T) \circ \log$$

$$\begin{aligned} \log \circ (\exp \circ T \circ \log) &= \sum \frac{(-1)^{n+1}}{n} (\exp \circ T \circ \log - Id)^n \\ &= \sum \frac{(-1)^{n+1}}{n} (\exp \circ (T - Id)^n \circ \log) \\ &= \exp \circ \left(\sum \frac{(-1)^{n+1}}{n} (T - Id)^n \right) \circ \log \\ &= \exp \circ \log(T) \circ \log \end{aligned}$$

Taking $T = Ad_\Lambda$ gives the result.

Exercise 9.11

i) $\hat{g}$ is the composition $r_g \circ l_{g^{-1}}$, both of which are analytic hence $\hat{g}$ is an analytic map so it is a morphism of G . Similarly, $Ad(g) = l_{g^{-1}} \circ r_g$ and each of r_g and $l_{g^{-1}}$ are analytic maps since $GL(L)$ is a p-adic analytic group. So $Ad(g)$ is a continuous map between p-adic analytic groups, hence it is an analytic map so it is a morphism between p-adic groups.

ii) Let

$$\Gamma = \{\gamma \in GL(L) : (\gamma - 1)(L) \subset pL\}$$

Then Γ is a uniform open subgroup of the p-adic group $GL(L)$ and

$$\mathcal{L}(GL(L)) = \mathbb{Q}_p \otimes L_\Gamma \cong \mathbb{Q}_p \otimes \log(\Gamma) = \mathfrak{gl}(L)$$

Let H be an open subgroup of G which is a uniform pro-p group such that $Ad(H) \subset \Gamma$. Then Ad induces

$$Ad^* : L_H \rightarrow \log(\Gamma) \quad Ad^* = \log \circ Ad|_{L_H}$$

From this we see that

$$\mathcal{L}(Ad) = Id \otimes Ad^* : \mathbb{Q}_p \otimes L_H = L \rightarrow \mathbb{Q}_p \otimes \log \Gamma = \mathfrak{gl}(L)$$

By 10.3, $Ad^* = ad$ and so

$$\mathcal{L}(Ad) = (Ad^*)^* = (ad)^*$$

Now, for $q, k \in \mathbb{Q}_p, h, g \in L_H$

$$(ad)^*(q \otimes h)(k \otimes g) = (Id_{\mathbb{Q}_p} \otimes ad)(q \otimes h)(k \otimes g) = (q Id_{\mathbb{Q}_p} \otimes ad(h))(k \otimes g) = qk \otimes ad(h)(g)$$

There are canonical isomorphisms

$$\begin{aligned} \alpha : \mathbb{Q}_p \otimes L_H &\xrightarrow{\sim} L & q \otimes h &\mapsto qh \\ \beta : \mathbb{Q}_p \otimes \log \Gamma &\xrightarrow{\sim} \mathfrak{gl}(L) & q \otimes A &\mapsto qA \end{aligned}$$

With respect to these associations

$$ad(qh)(kg) = (qh, kg) = (ad)^*(q \otimes h)(k \otimes g)$$

where the first ad is a map $L \rightarrow \mathfrak{gl}(L)$. And by the functoriality of the Lie functor

$$\mathcal{L}(Ad) = \beta \circ (Id_{\mathbb{Q}_p} \otimes Ad^*) \circ \alpha^{-1}$$

so we see that

$$ad = \beta \circ (Id_{\mathbb{Q}_p} \otimes Ad^*) \circ \alpha^{-1}$$

hence

$$\mathcal{L}(Ad) = ad$$

as desired.

Exercise 9.12

Let $K = \ker Ad$ where $Ad : G \rightarrow GL(L)$ where $L = \mathcal{L}(G)$. Since $GL(L) \cong GL_d(\mathbb{Q}_p)$, G/K is isomorphic to a subgroup of $GL_d(\mathbb{Q}_p)$. Moreover, $K = Ad^{-1}(Id)$ is a closed subset of G since Ad is continuous and $\{Id\}$ is closed in $GL(L)$. Hence $K \leq_c G$. Then, let G^+ be the subspace generated by all uniform open subgroups of G . Then G^+ is open in G so G/G^+ is discrete. Now we show $K = C_G(G^+)$. Let $x \in K$ then $Ad(x) = I$ so $x \in C_G(G^+)$. Let $x \in C_G(G^+)$ and H be a uniform open subgroup of G . Then $L = \mathbb{Q}_p \otimes L_H$ and, associating $\mathbb{Q}_p \otimes L_H$ with $\mathbb{Q}_p L_H$, we can write

$$Ad(x)(qh) = x(qh)x^{-1} = q(xhx^{-1}) = qh$$

so $Ad(x) = Id$ hence $x \in K$. Thus $K = C_G(G^+)$. Now put $Z = K \cap G^+$, then Z is abelian and $K/Z \cong KG^+/G^+ \leq G/G^+$ so K/Z is discrete. Z is normal because for $g \in G$, $gKg^{-1} = K$ and a conjugate of a uniform open subgroup is also open since conjugation is a homeomorphism so $gG^+g^{-1} = G^+$

so $gZg^{-1} = Z$. And finally Z is closed in G since K is closed and so is G^+ since each of the generator uniform open subgroups H are also closed in G .

Note: We needed to take the subgroup generated by all uniform open subgroups of G merely to make Z normal in G . Notice that $C_G(G^+) = C_G(H) = K$ for any uniform open subgroup H of G .

0 Questions

Exercise 2.2

For part (ii), I do not know how to prove $[N^p, M] \geq [N, M]^p$.

Exercise 6.3

Let A be a normed $\mathbb{Q}_p$ -algebra and V a finite dimensional vector subspace of A . The inclusion map $\iota : V \rightarrow A$ is continuous with respect to the p-adic topology on V and the norm topology on A . I do not know whether the latter topology always induces the former topology on V . The guess is NO because A might not be complete with respect to the norm topology but V is always complete with respect to the p-adic topology. (Why?)

Exercise 10.3

I do not know how to prove $Ad_\Lambda(g) \in \Delta$. It has something to do with the Haussdorf-Campbell formula to estimate the difference $\log(gug^{-1}) - \log u$ which is given by $\log(g^{-1}, u^{-1}) + \dots$

Chapter 9.5

How to actually conceptualize $\mathcal{L}(G)$ as the tangent space at the identity of the p-adic group G ?

General

What is the right way to think about uniform pro-p groups and the notion of distance in these groups? What does that have to do with being able to extend the norm on $\mathbb{Z}_p[G]$ to $\mathbb{Q}_p[G]$?

Exercise 8.7

I do not know how to show that $GL_n(\mathbb{Z}_p)$ is a maximal compact subgroup of $GL_n(\mathbb{Q}_p)$?

Exercise 8.4

To be solved.

Exercise 9.5

To be written.

Exercise 9.8

To be written.

0.1 Question

A representation $\rho : \widehat{SL_n(\mathbb{Z})} \rightarrow GL(V)$ is continuous if and only if it factors through a finite group F ?